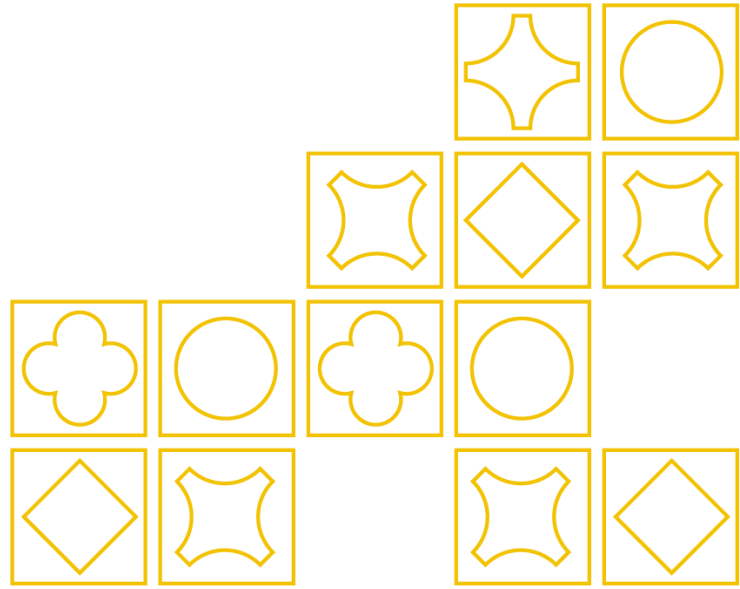
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AI Essentials for Academic Success



Today's Session

Welcome and Introductions
Professor Truman Academic Director of First-Year Seminars
Professor Syta

1 AI Tools for Academic Success

Professor Kousen

Visiting Professor of the Practice, Computer Science

Associate Director for STEM Initiatives, Elting Center for Innovation and Entrepreneurship

2 AI and Academic Success

Professor Syta

Charles A. Dana Research Associate Professor and Chair of Computer Science

Faculty Lead, Elting Center for Innovation and Entrepreneurship

3 AI Essentials Lab

Dr. Purrington

AI Academic Fellow

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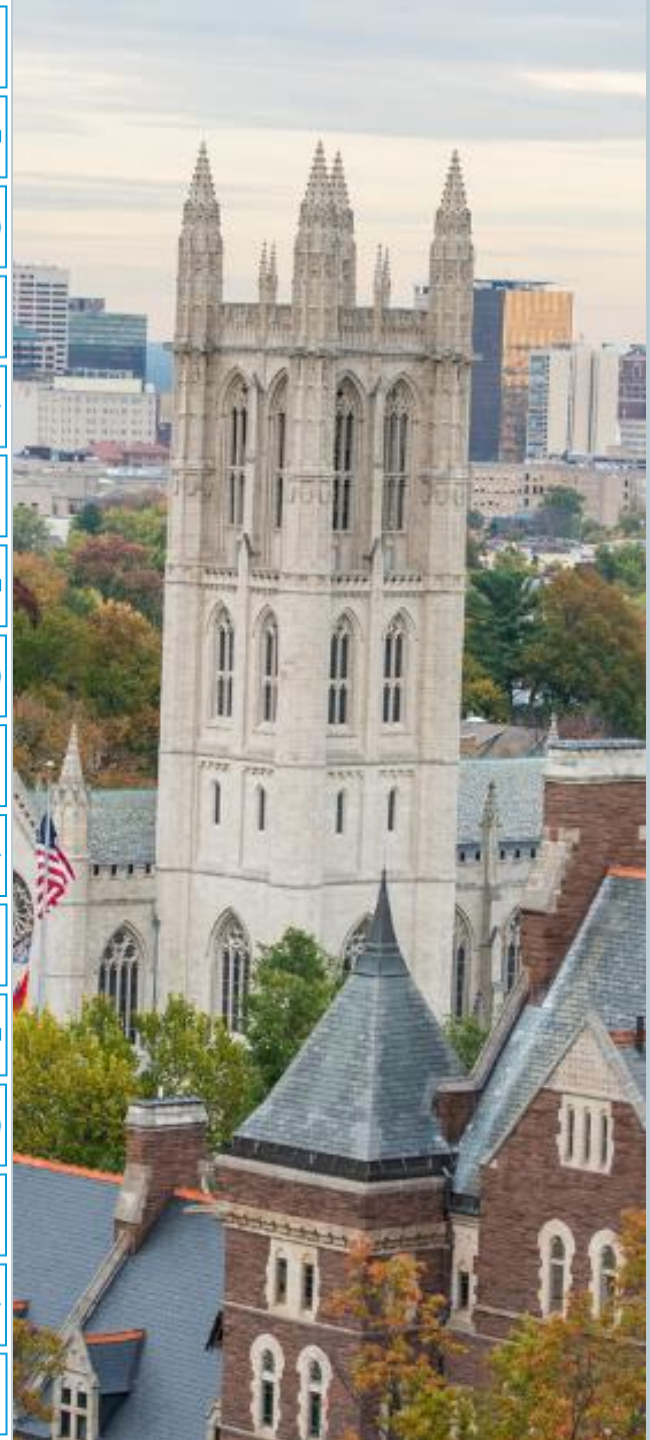
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Prof. James Truman

Academic Director of First-Year Seminars

- **“Trinity College prepares students to be bold, independent thinkers who lead transformative lives.”**
- The First-Year Seminar is your invitation to the intellectual work that is the foundation of the liberal arts
- Discussion and writing are how you develop your judgement, your taste, and your critical thinking skills



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Liberal Arts in the Age of AI

NOT AI makes a liberal arts education less relevant

Instead: There has never been a better time to be at Trinity

- Liberal arts education builds durable skills
  - Critical thinking, problem solving, communication, and more
- AI can give you answers but it is all about the questions
- There is no prompt for durable skills

Takeaway: Liberal arts education builds durable skills and the judgment to choose the questions, test the answers, and decide what to do with them, regardless of the tools available.

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Where We're Going

NOT A session about whether AI is good or bad

Instead: A starting point for your own judgment

- We are not here to tell you what to think about AI
- We are here to help you
  - You need knowledge, skills, and judgment to navigate college, and the world after it
  - In an environment where AI is everywhere

Takeaway: This session not about AI. It is about your academic success.

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AI Is Already Part of Your Environment

NOT "I use AI" vs. "I don't use AI"

Instead: You are already using it, and you will be asked to use it - know it

- Built in already
  - Search results, Copilot with everyday tools
- Unlabeled all along
  - Translation, recommendations and spam filters have run on AI for years
- Expected of you soon
  - Some courses might require it, some internships and employers already do

Takeaway: Opting out entirely is no longer a real option. You must learn how AI works, what it does well, where it fails, and whether, when, and how to use it.

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AI Literacy: Develop Your Skills & Position

NOT We will tell you what you should think about AI

Instead: Wherever you start from, you should be able to say why. And re-examine.

- Enthusiastic, deeply skeptical, somewhere in between, or simply tired of the conversation
  - Fine places to start, none is a position until you can say why
  - And you need to re-examine your position as things change
- Developing your judgment is your responsibility
- Helping you develop it is part of ours

Takeaway: Keep Asking
What do I actually understand about AI? What reasons support the choices I make?

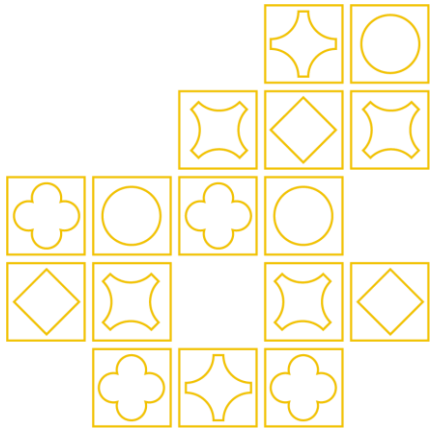
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AI Literacy Is *SO* Much More Than ChatGPT

NOT Learn the “right prompts”

Instead: Develop AI literacy.

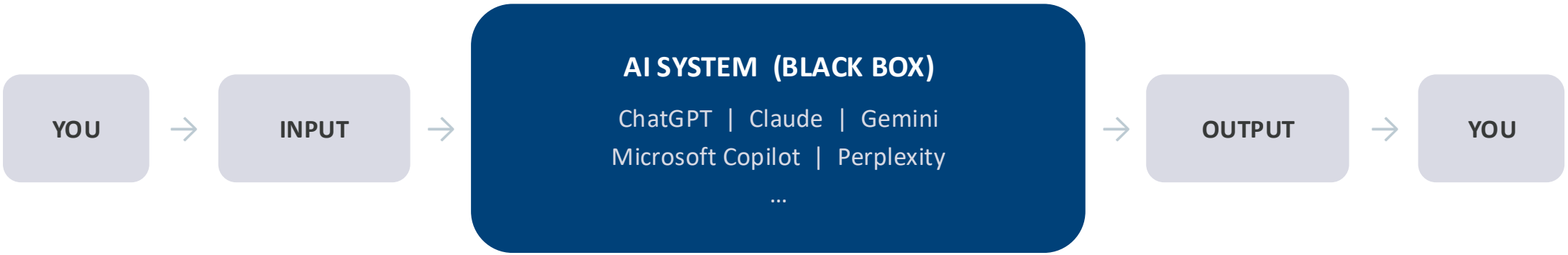
- Prompting is the smallest part of it
- *SO* much more to learn!



Takeaway: Tool proficiency matters. AI literacy is required.

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Using AI Starts and Ends with You



Takeaway: **Every stage is a decision you make. Your judgment must run through the whole process.**

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Before You Ask: Reasoning About Input



PURPOSE

Should I be asking AI at all?

- Is this the right tool for what I am trying to do?
- Would an answer here skip thinking I need to do myself?
- Is this a question AI should be answering?

PROMPT AND CONTEXT

How am I asking, and what does it not know?

- The prompt is how you ask. The context is what it has to work with.
- Whatever you leave out, it fills in with something plausible.

PRIVACY

What am I putting in?

- Is this mine to share?
- Consumer tools may keep it and train on it. Licensed tools usually cannot. Know which one you are in.

INFLUENCE

What will the answer do to me?

- It arrives before you have formed your own view.
- It frames the problem and pulls your thinking with it.
- Sometimes that helps. Sometimes you need to think first.

Takeaway: Judgment starts before the prompt.

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After It Answers: Reasoning About Output



ACCURACY

Is it true?

- Fluent, confident, and wrong is a normal failure mode.
- Invented facts, sources, and quotations.
- Check it!
- Cannot verify it? Do not use it.

BIAS

Whose framing is this?

- The answer reflects its training data, gaps and biases included.
- You get one framing, and it rarely says what it left out.
- Push back and it will often just agree. That is not evidence you were right.
- Ask what someone who disagrees would say.

INTEGRITY

What do I do with it?

- Use it, revise it, or discard it. Decide.
- If you keep it, say where it came from.
- Can I defend the result as my own work?

ACCOUNTABILITY

Who is responsible for it?

- Insecure code, unsafe advice, wrong instructions: the consequences are real.
- Not the AI. Not the company that built it.
- You submitted it. You acted on it.

Takeaway: Judgment does not end when the answer arrives.

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Inside the Box: Reasoning About AI



OPACITY

Why can I not see how it got there?

- You see what goes in and what comes out. The steps between are not visible.
- It can give you an explanation, but that is another output, not a record of its work.

VARIABILITY

Will I get the same answer twice?

- The same question can produce a different answer each time.
- It predicts likely text. Even when it searches, the wording is generated, not quoted.

DIFFERENCES

Do these systems all work the same way?

- ChatGPT, Claude, Gemini and Copilot are built and tuned differently.
- When two systems disagree, that is your signal to check.

CHANGE

How long will what I know about this system stay true?

- Models are updated, replaced and retired without notice.
- Judge the system in front of you, not the one you used last time.

Takeaway: These systems should be explainable. They are not. Judge accordingly.

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Start With the Learning, Not the Tool

NOT Can I use AI for this?

Instead: Ask what the assignment is for, then ask about the tool.

- What is this supposed to teach me?
- Would AI support that learning, or bypass it?
- What should I be able to do unaided by the end?
- Of course, what you can do depends on the policy for that course

- But why professors assign work? And grade?

Takeaway: This makes AI use a learning decision, not just a compliance decision.

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Academic Integrity **Matters**

NOT A rule you follow so you do not get caught

Instead: Taking full responsibility for the work you submit.

- Do the work or give others credit
- Know what counts as dishonesty: cheating, plagiarism, unauthorized collaboration, falsification, misrepresenting authorship, submitting the same work twice
- Ask when you are unsure: your professor first and for general questions, but Trinity's research librarians are a great resource too
- **None of this is only academic: it is practice for how you work and lead after Trinity**

Takeaway: Integrity protects what you came here for:
learning that is yours and work others can trust and depend on.

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Your **Professor** Sets the Policy

NOT Trinity has one universal rule for AI use in every course

Instead: Each professor sets the policy for their course and sets it for a reason.

- Where: Find it in the syllabus and follow it
- Why: Your professors are experts in their fields, they know what their course is meant to build in you, and they have thought hard about what AI supports and what it quietly replaces
- Your part: Follow it but understand and ask

Takeaway: Trust their expertise but ask.
The policy tells you the boundary but understanding the reason builds your judgment.

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Different Classes, Different Rules, for a Reason

A professor may reasonably say:

YES	NO	SOMETIMES
with attribution and disclosure	not in this course	permitted, but not on this assignment

Always read the syllabus, check the instructions on each assignment as the answer can differ, never assume

Takeaway: Ask before you used it and not after you submit it.

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If You Use AI, Disclose It and Keep the Record

DISCLOSE

Say what you used, and how

- Name the tool and the date.
- Say what you used it for, and how you verified the result.
- Cite AI sources the way your professor asks.

DOCUMENT

Keep the process, not just the output

- Save your prompts, or the transcript of the exchange.
- Record the tool's name and version.
- If your use is questioned, that record is your defense.

RESPECT OTHERS' WORK

Do not feed it what is not yours

- Never put course materials (slides, problem sets, readings, etc.) into AI without consent.
- They are proprietary and often copyrighted.

AI MISUSE

It falls under existing rules

- Unauthorized collaboration: using AI where it was prohibited.
- Misrepresenting authorship: submitting AI work undisclosed.
- Falsification: you answer for what you submit.

Takeaway: Do your own work and fully credit the work of others.

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Don't Replace the **Humans** in Your Education

When you are stuck or unsure or things get hard, there are many **people** to go to:

Your professors
Office hours.

Research librarians
They find what you cannot.

Classmates
Your network.

Tutors, TAs, others
A resource worth using early.

AI is **always** available and never busy. What it cannot give you is **mentorship**, **productive disagreement**, **collaboration**, or someone who knows your **field** and knows **you**. And a **lifelong** relationship.

Takeaway: Do These relationships are not an inefficiency in your education. They are your education.

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The AI Studio, at the Gates Quad entrance to Raether Library, has AI help for students.

AI and **Your** Academic Success

THE TASK

Completing the task ≠ learning from the task

- Complete, correct, on time with nothing learned.
- AI can produce the artifact without producing the learning.
- The bill comes due at the exam, in seminar, in the next course.

THE STRUGGLE

Don't automatically eliminate the struggle

- Finding it hard is part of learning, not a sign you are doing it wrong.
- Ask: am I stuck, or just uncomfortable?
- Sometimes AI is right for getting unstuck. Knowing which is the skill.

BETTER ASKS

Use AI to increase your thinking

- Quiz me. A hint, not the answer.
- Challenge my argument. What am I assuming?
- Show me what I still do not understand.

OTHER TOOLS

Sometimes the best use of AI is no AI

- Textbook, original source, database, documentation, or even a calculator.
- Your own reasoning, given more time.
- Or a person who knows the answer, and why.

Takeaway: After using AI, have I done more thinking — or less?

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Don’t Form Your Views Based on Today’s AI

2018 — GPT-1
Research model
- Language-model research
- Demonstrates generative pretraining approach

2020 — GPT-3
Scaling changes what models can do
- Much larger model
- Striking few-shot capabilities
- General-purpose language generation becomes much more compelling

2022 — ChatGPT
AI becomes conversational—and mainstream
- Natural speech
- Accessible to everyone
- Generative AI moves rapidly into everyday life

2023 — GPT-4
Stronger capability + multimodality
- More capable reasoning
- Increasing ability to work across modalities
- AI becomes useful for substantially more complex work

2025–26 — Reasoning + Agents
From answering → reasoning → acting
- More computation devoted to difficult problems
- Much larger context
- Tool use

When you started high school, ChatGPT did not exist. Now you are starting college in the era of agentic AI.

Things change

Takeaway: AI literacy is a process, not a destination.

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Your Academic Success Is **Yours**

NOT There is one right way to succeed in college

Instead: Your education is something you actively build.

- Your professors will teach, challenge, guide, and support you
- You will have access to tools
- But you decide how to utilize those
- AI can help
- You do not have to figure everything out on your own
- Ask questions and use resources around you

Takeaway: Your academic success is yours, but you are not alone.
We are here to help you build it.

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3 AI Essentials Lab

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AI Essentials Lab

- Orientation
- AI literacy course
- Hands-on workshop
- Symposium

Scan for details

trincoll.edu/academic-ai/

Bold, independent thinkers. Informed judgment.

Thank YOU!

Scan the QR code to take the survey!

Tell us how we did!

Bold, independent thinkers. Informed judgment.